# 마이크로프로세서

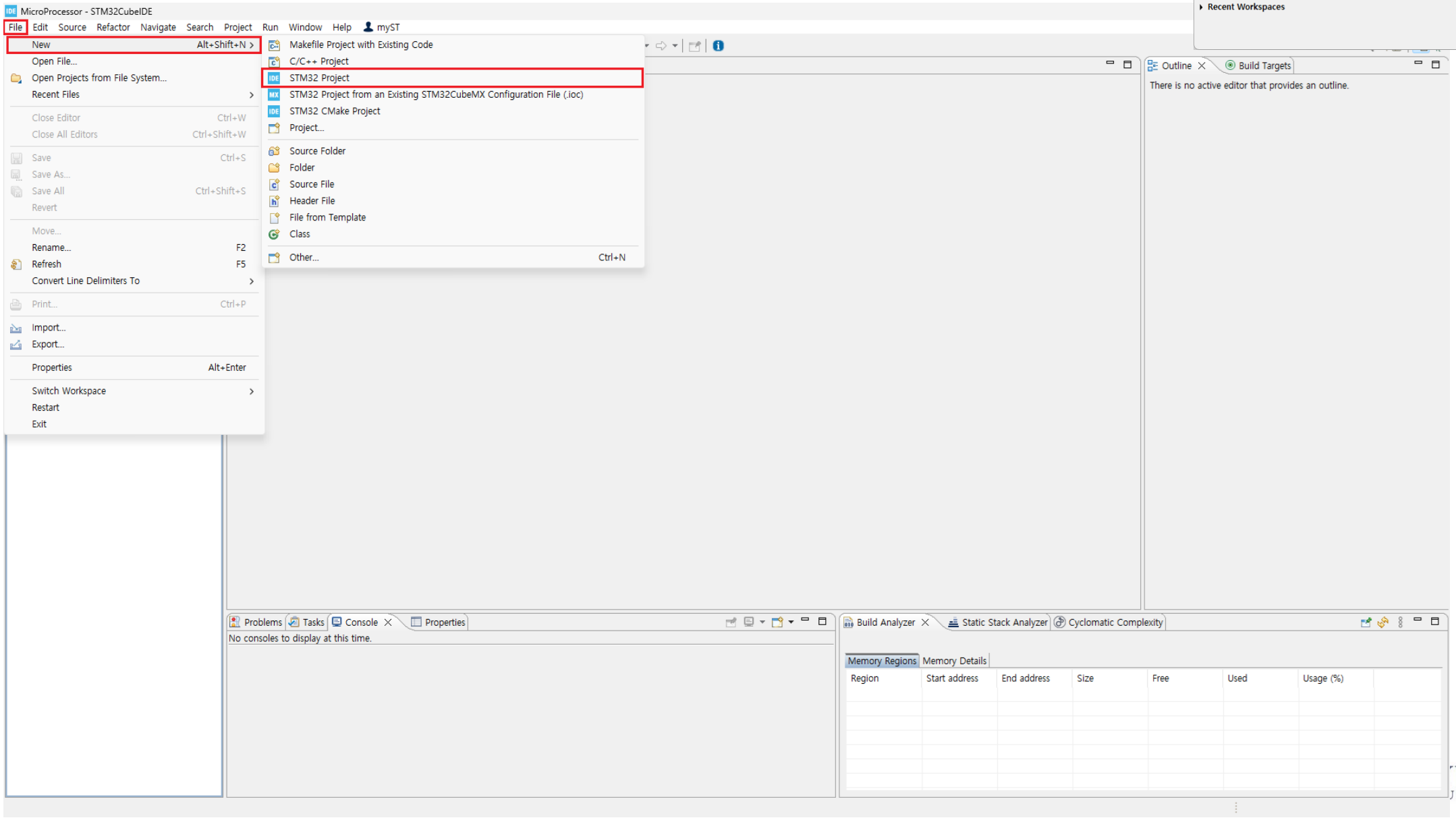
## Chapter 2. Getting Start with STM32

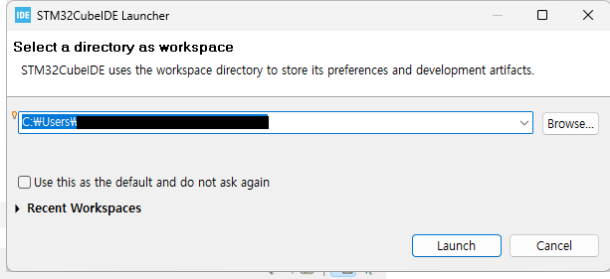
Human-Robot Interaction Laboratory

KYUNG HEE UNIVERSITY

# STM32CudeIDE Project

## >> STM32CubeIDE

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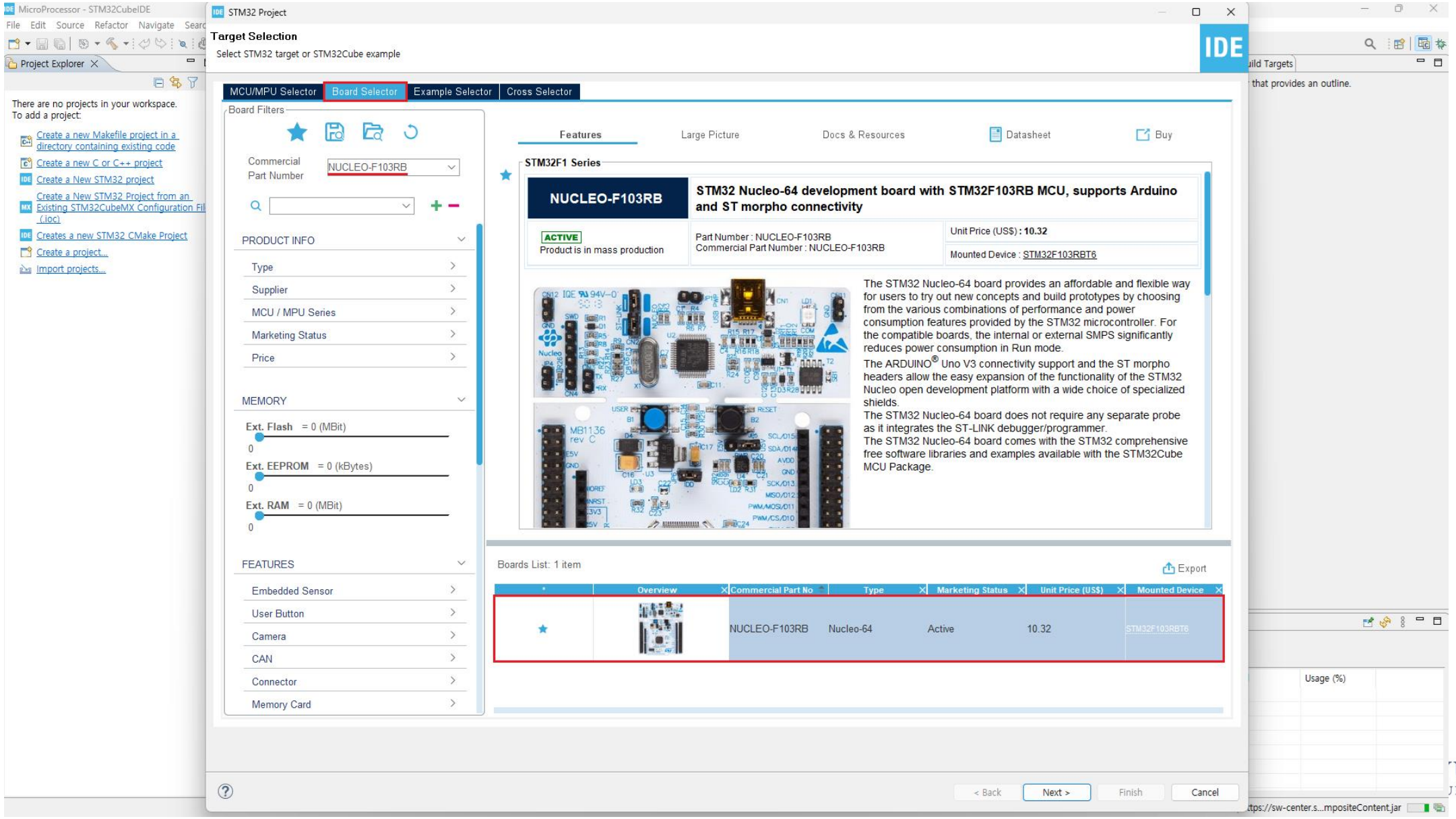
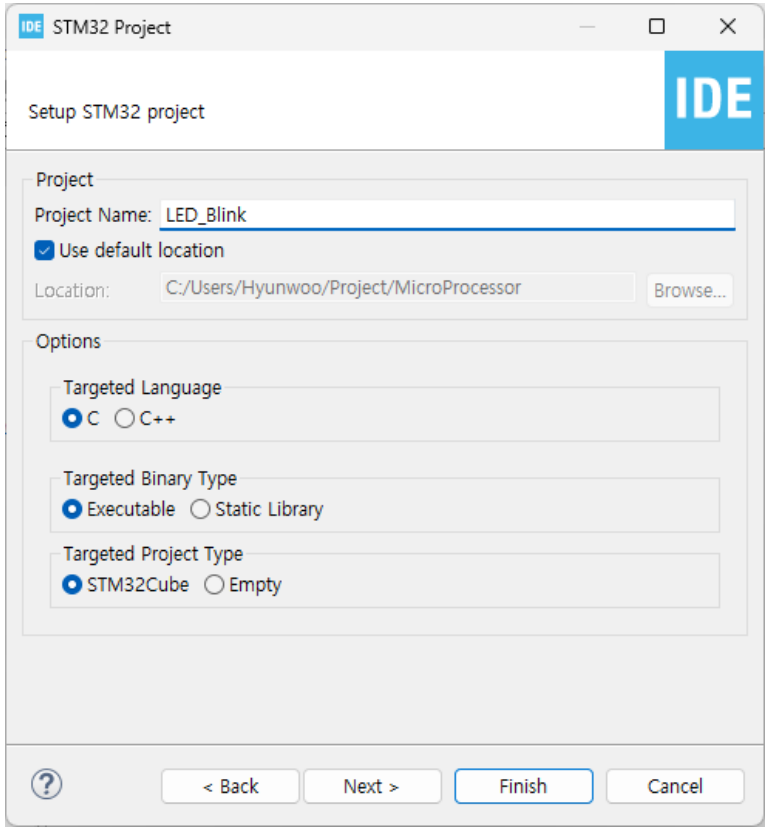
# STM32CudeIDE Project

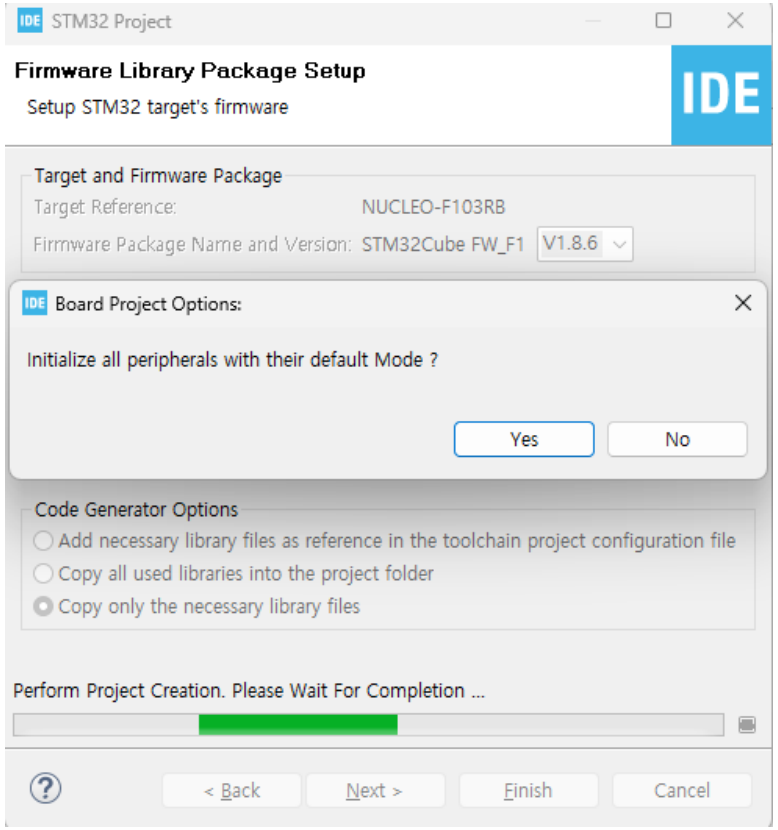
## >> STM32CubeIDE

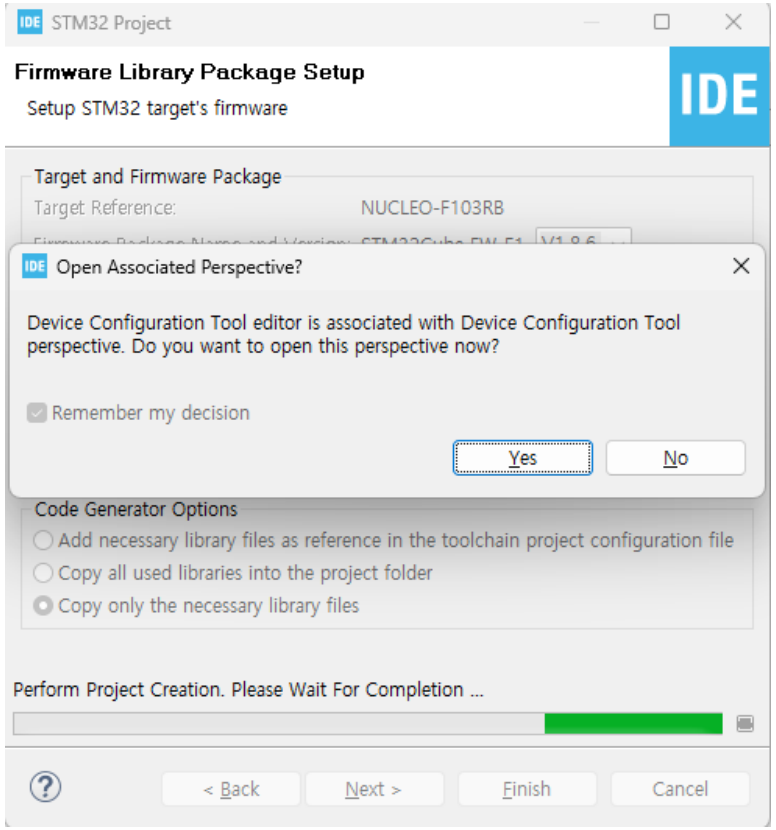


# STM32CudeIDE Project

## >> STM32CubeIDE

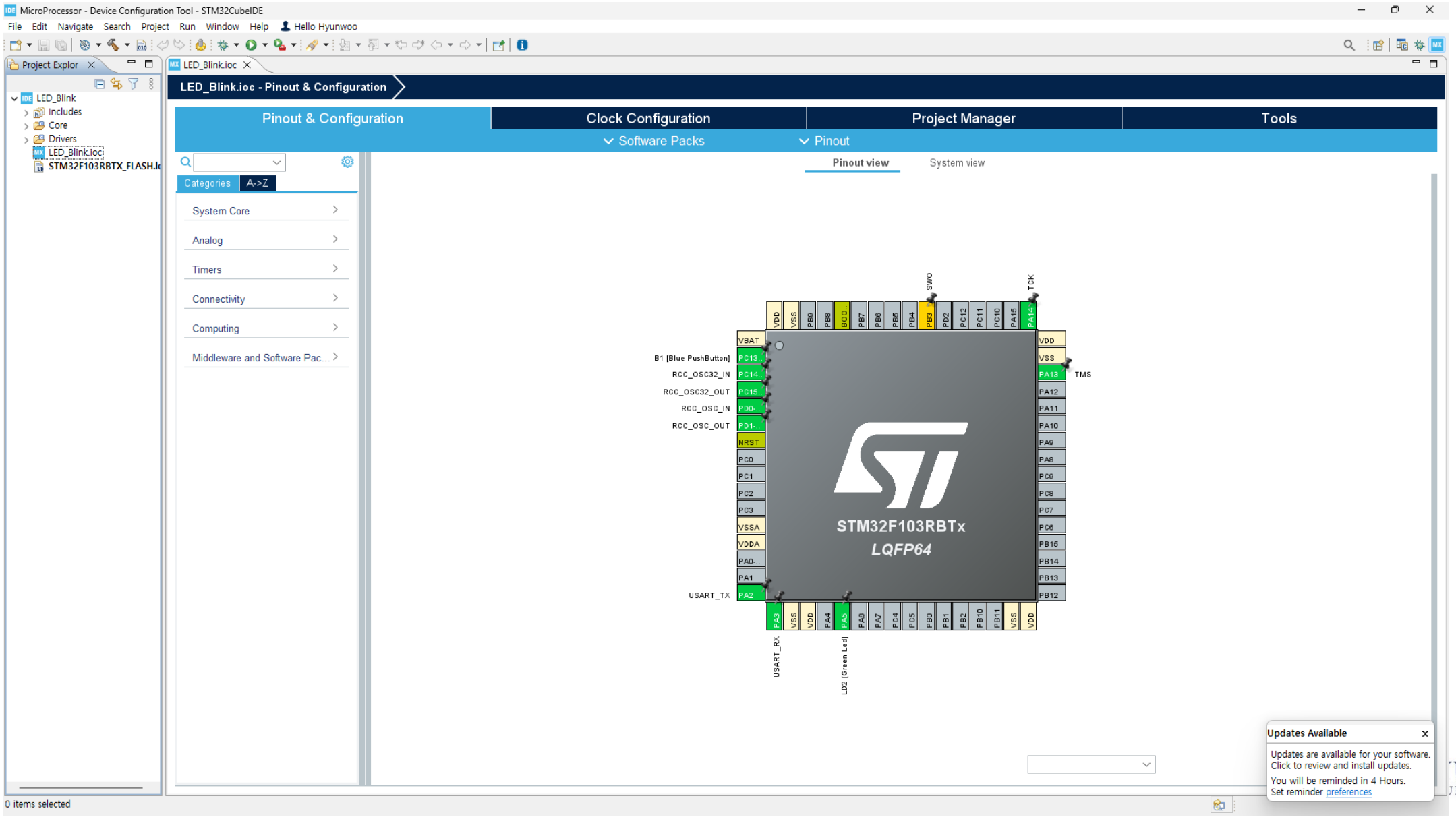


# STM32CudeIDE Project

## >> LED_Blink

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# STM32CudeIDE Project

## >> LED_Blink

MicroProcessor - Device Configuration Tool - STM32CubeIDE

File Edit Navigate Search Project Run Window Help Hello Hyunwoo

Project Explor

- LED_Blink
  - Includes
  - Core
  - Drivers
  - LED_Blink.ioc
  - STM32F103RBTX_FLASH.l

*LED_Blink.ioc

LED_Blink.ioc - Pinout & Configuration

Pinout & Configuration | Clock Configuration | Project Manager | Tools

Software Packs | Pinout

Categories | A->Z

System Core
- DMA
- GPIO
- IWDG
- NVIC
- RCC
- SYS
- WWDG

Analog

Timers

Connectivity

Computing

Middleware and S...

RCC Mode and Configuration

Mode

High Speed Clock (HSE) Disable

Low Speed Clock (LSE) Disable

Master Clock Output

Configuration

Reset Configuration

Parameter Settings | User Constants | NVIC Settings

Configure the below parameters :

Search (Ctrl+F)

System Parameters

| | |
|---|---|
| VDD voltage (V) | 3.3 V |
| Prefetch Buffer | Enabled |
| Flash Latency(WS) | 2 WS (3 CPU cycle) |

RCC Parameters

| | |
|---|---|
| HSI Calibration Value | 16 |
| HSE Startup Timout Value (ms) | 100 |
| LSE Startup Timout Value (ms) | 5000 |

Pinout view | System view

STM32F103RBTx

LQFP64

B1 [Blue PushButton] (PC13..)
RCC_OSC32_IN (PC14..)
RCC_OSC32_OUT (PC15..)
RCC_OSC_IN (PD0-..)
RCC_OSC_OUT (PD1-..)
USART_TX (PA2)
USART_RX (PA3)
LD2 [Green Led] (PA5)
SWO (PB3)
TCK (PA14)
TMS (PA13)

Updates Available

Updates are available for your software.
Click to review and install updates.
You will be reminded in 4 Hours.
Set reminder preferences

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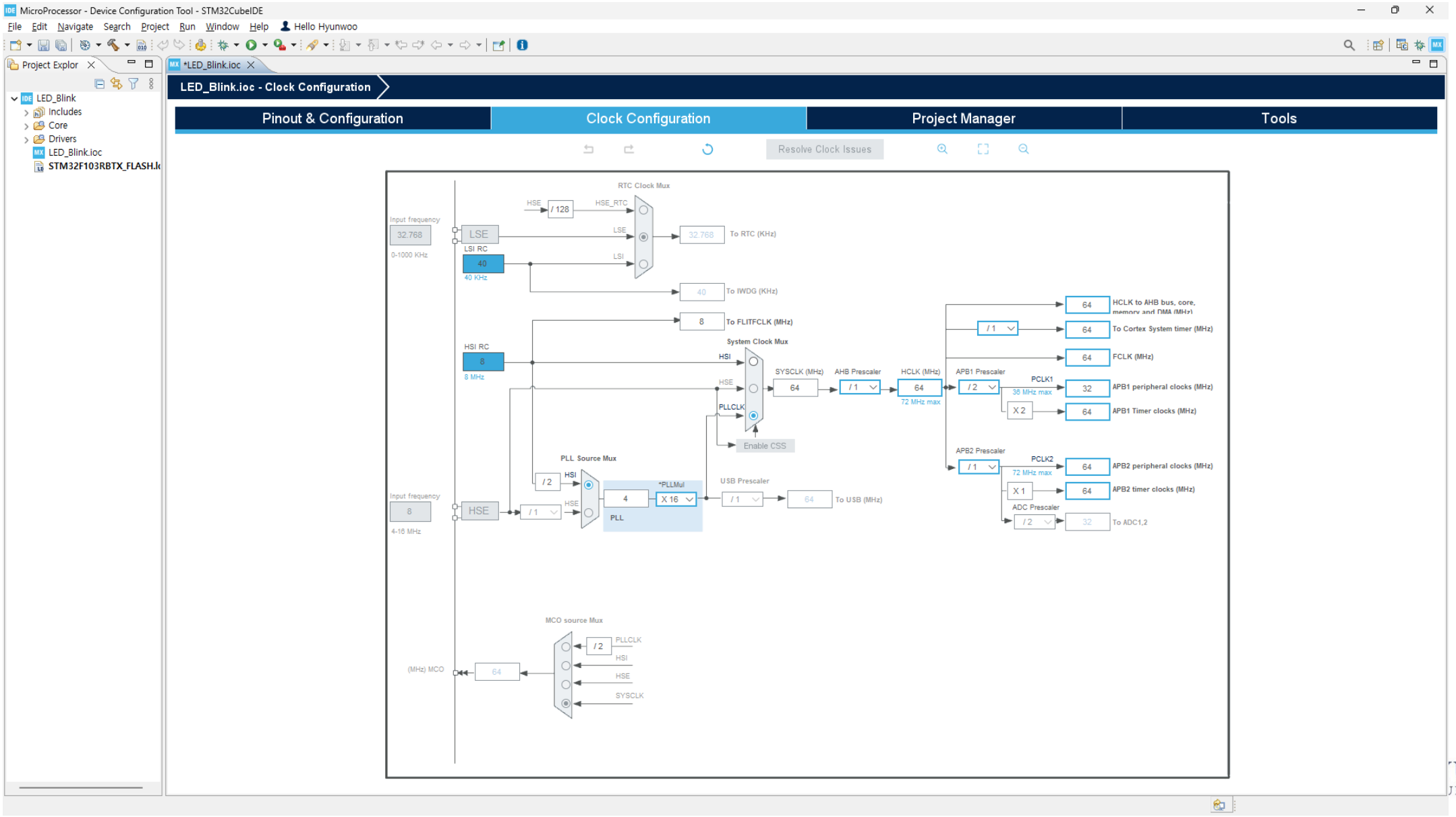
# STM32CudeIDE Project

## >> LED_Blink

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# STM32CudeIDE Project

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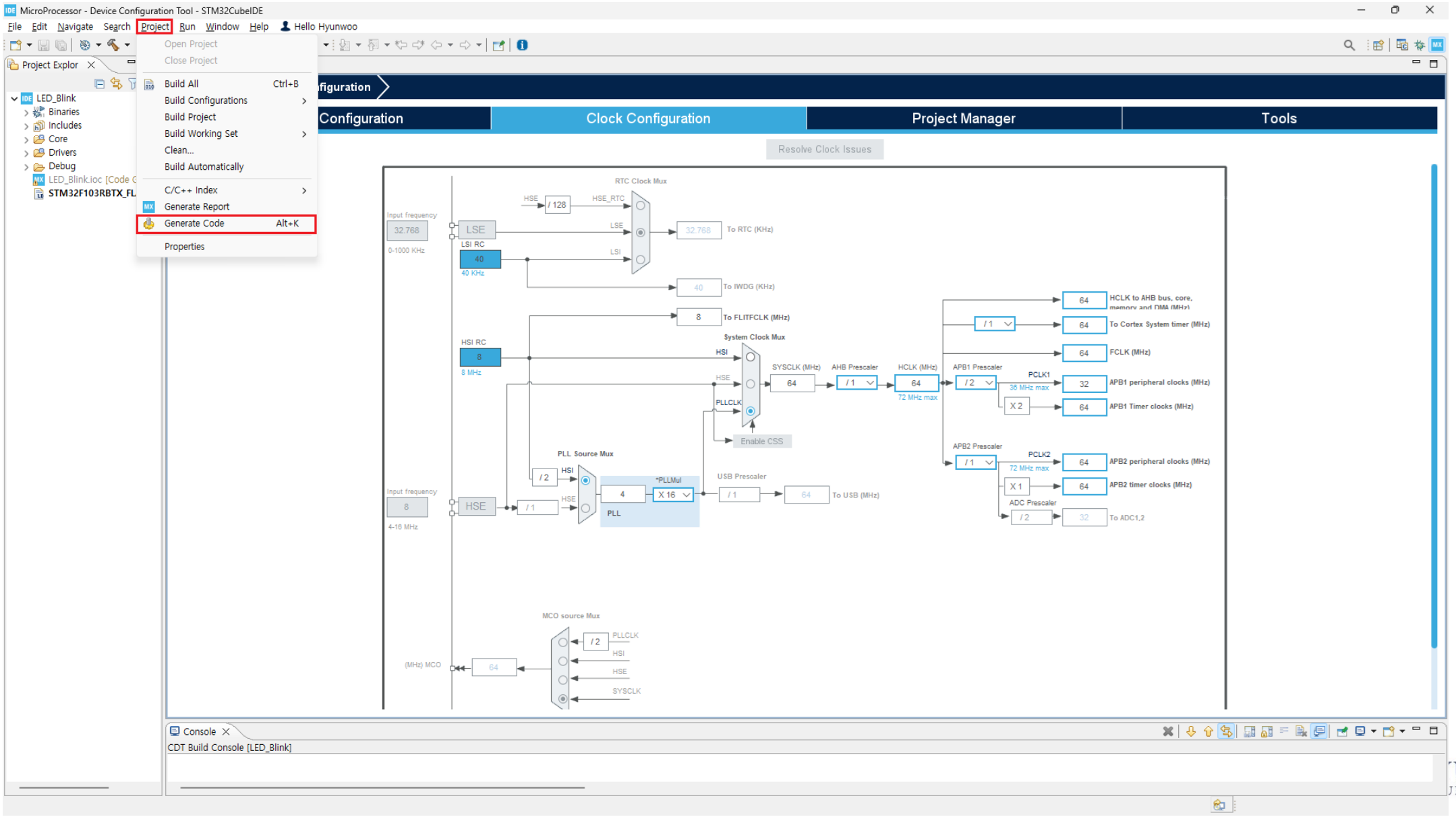

# STM32CudeIDE Project

## >> LED_Blink

MicroProcessor - LED_Blink/Core/Src/main.c - STM32CubeIDE

File Edit Source Refactor Navigate Search Project Run Window Help Hello Hyunwoo

Project Explorer

- LED_Blink
  - Binaries
  - Includes
  - Core
  - Drivers
  - Debug
  - LED_Blink.ioc
  - STM32F103RBTX_FLASH.ld

LED_Blink.ioc | main.c

```c
/* USER CODE BEGIN Header */
/**
  ******************************************************************************
  * @file           : main.c
  * @brief          : Main program body
  ******************************************************************************
  * @attention
  *
  * Copyright (c) 2025 STMicroelectronics.
  * All rights reserved.
  *
  * This software is licensed under terms that can be found in the LICENSE file
  * in the root directory of this software component.
  * If no LICENSE file comes with this software, it is provided AS-IS.
  *
  ******************************************************************************
  */
/* USER CODE END Header */
/* Includes ------------------------------------------------------------------*/
#include "main.h"

/* Private includes ----------------------------------------------------------*/
/* USER CODE BEGIN Includes */

/* USER CODE END Includes */

/* Private typedef -----------------------------------------------------------*/
/* USER CODE BEGIN PTD */

/* USER CODE END PTD */

/* Private define ------------------------------------------------------------*/
/* USER CODE BEGIN PD */

/* USER CODE END PD */

/* Private macro -------------------------------------------------------------*/
/* USER CODE BEGIN PM */

/* USER CODE END PM */
```

Outline | Build Targets

- main.h
- huart2 : UART_HandleTypeDef
- SystemClock_Config(void) : void
- MX_GPIO_Init(void) : void
- MX_USART2_UART_Init(void) : void
- main(void) : int
- SystemClock_Config(void) : void
- MX_USART2_UART_Init(void) : void
- MX_GPIO_Init(void) : void
- Error_Handler(void) : void
- assert_failed(uint8_t*, uint32_t) : void

Problems | Tasks | Console | Properties

CDT Build Console [LED_Blink]

```
arm-none-eabi-objdump -h -S LED_Blink.elf  > "LED_Blink.list"
   text    data     bss     dec     hex filename
   5536      12    1644    7192    1c18 LED_Blink.elf
Finished building: default.size.stdout

Finished building: LED_Blink.list


13:10:07 Build Finished. 0 errors, 0 warnings. (took 2s.657ms)
```

Build Analyzer | Static Stack Analyzer | Cyclomatic Complexity

Memory Regions | Memory Details

| Region | Start address | End address | Size | Free | Used | Usage (%) |
|---|---|---|---|---|---|---|

Writable | Smart Insert | 1 : 1 : 0

# STM32CudeIDE Project

## >> LED_Blink

MicroProcessor - LED_Blink/Core/Src/main.c - STM32CubeIDE

File Edit Source Refactor Navigate Search Project Run Window Help Hello Hyunwoo

Open Project
Close Project
Build All Ctrl+B
Build Configurations
Build Project
Build Working Set
Clean...
Build Automatically
Build Targets
C/C++ Index
Generate Report
Generate Code
Properties

Project Explorer
- LED_Blink
  - Binaries
  - Includes
  - Core
  - Drivers
  - Debug
  - LED_Blink.ioc
  - STM32F103RBTX_FLASH.ld

```
  */
/* USER CODE END Header */
/* Includes ------------------------------------------------------------------*/
#include "main.h"

/* Private includes ----------------------------------------------------------*/
/* USER CODE BEGIN Includes */

/* USER CODE END Includes */

/* Private typedef -----------------------------------------------------------*/
/* USER CODE BEGIN PTD */

/* USER CODE END PTD */

/* Private define ------------------------------------------------------------*/
/* USER CODE BEGIN PD */

/* USER CODE END PD */

/* Private macro -------------------------------------------------------------*/
/* USER CODE BEGIN PM */

/* USER CODE END PM */
```

Outline / Build Targets
- main.h
- huart2 : UART_HandleTypeDef
- SystemClock_Config(void) : void
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- main(void) : int
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- MX_GPIO_Init(void) : void
- Error_Handler(void) : void
- assert_failed(uint8_t*, uint32_t) : void

Problems Tasks Console Properties

CDT Build Console [LED_Blink]

```
arm-none-eabi-objdump -h -S LED_Blink.elf  > "LED_Blink.list"
   text    data     bss     dec     hex filename
   5536      12    1644    7192    1c18 LED_Blink.elf
Finished building: default.size.stdout

Finished building: LED_Blink.list


13:12:34 Build Finished. 0 errors, 0 warnings. (took 555ms)
```

Build Analyzer / Static Stack Analyzer / Cyclomatic Complexity

Memory Regions / Memory Details

| Region | Start address | End address | Size | Free | Used | Usage (%) |
|---|---|---|---|---|---|---|

Writable | Smart Insert | 1 : 1 : 0

# STM32CudeIDE Project

## >> STM32CubeIDE

- CubeIDE는 STM32 마이크로컨트롤러/프로세서 개발을 위한 통합 개발 환경
- GCC 툴체인과 GDB 디버깅 기능(레지스터, 메모리, 변수, 중단점 등 실시간 고급 디버깅 기능)
- 프로세서의 기본적인 설정, Pinout 등의 설정은 GUI(CubeMX)상에서 쉽게 설정이 가능함
- 핀아웃 및 클럭 설정 충돌 자동 해결 기능
- CubeIDE와 CubeMX와 연동되어 자동으로 C 코드 프로젝트 생성
- CubeIDE 이외의 IDE 툴체인(IAR, KEIL 등)과 호환 가능한 C 코드 생성

# STM32CudeIDE Project

## >> LED_Blink – stm32f1xx_hal_gpio.h

MicroProcessor - LED_Blink/Drivers/STM32F1xx_HAL_Driver/Src/stm32f1xx_hal_gpio.c - STM32CubeIDE

File Edit Source Refactor Navigate Search Project Run Window Help Hello Hyunwoo

Project Explorer

- LED_Blink
  - Binaries
  - Includes
  - Core
  - Drivers
    - CMSIS
    - STM32F1xx_HAL_Driver
      - Inc
      - Src
        - stm32f1xx_hal_cortex.c
        - stm32f1xx_hal_dma.c
        - stm32f1xx_hal_exti.c
        - stm32f1xx_hal_flash_ex.c
        - stm32f1xx_hal_flash.c
        - stm32f1xx_hal_gpio_ex.c
        - stm32f1xx_hal_gpio.c
        - stm32f1xx_hal_pwr.c
        - stm32f1xx_hal_rcc_ex.c
        - stm32f1xx_hal_rcc.c
        - stm32f1xx_hal_tim_ex.c
        - stm32f1xx_hal_tim.c
        - stm32f1xx_hal_uart.c
        - stm32f1xx_hal.c
      - LICENSE.txt
  - Debug
  - LED_Blink.ioc
  - STM32F103RBTX_FLASH.ld

LED_Blink.ioc | main.c | stm32f1xx_hal_gpio.c

```c
  }
}

/**
  * @brief  Toggles the specified GPIO pin
  * @param  GPIOx: where x can be (A..G depending on device used) to select the GPIO peripheral
  * @param  GPIO_Pin: Specifies the pins to be toggled.
  * @retval None
  */
void HAL_GPIO_TogglePin(GPIO_TypeDef *GPIOx, uint16_t GPIO_Pin)
{
  uint32_t odr;

  /* Check the parameters */
  assert_param(IS_GPIO_PIN(GPIO_Pin));

  /* get current Output Data Register value */
  odr = GPIOx->ODR;

  /* Set selected pins that were at low level, and reset ones that were high */
  GPIOx->BSRR = ((odr & GPIO_Pin) << GPIO_NUMBER) | (~odr & GPIO_Pin);
}

/**
* @brief  Locks GPIO Pins configuration registers.
* @note   The locking mechanism allows the IO configuration to be frozen. When the LOCK sequence
*         has been applied on a port bit, it is no longer possible to modify the value of the port bit until
*         the next reset.
* @param  GPIOx: where x can be (A..G depending on device used) to select the GPIO peripheral
* @param  GPIO_Pin: specifies the port bit to be locked.
*         This parameter can be any combination of GPIO_PIN_x where x can be (0..15).
* @retval None
*/
HAL_StatusTypeDef HAL_GPIO_LockPin(GPIO_TypeDef *GPIOx, uint16_t GPIO_Pin)
{
  __IO uint32_t tmp = GPIO_LCKR_LCKK;

  /* Check the parameters */
  assert_param(IS_GPIO_LOCK_INSTANCE(GPIOx));
  assert_param(IS_GPIO_PIN(GPIO_Pin));
```

**GPIO 핀 상태 반전(Toggle) 함수**

Outline | Build Targets

- stm32f1xx_hal.h
- GPIO_MODE
- EXTI_MODE
- GPIO_MODE_IT
- GPIO_MODE_EVT
- RISING_EDGE
- FALLING_EDGE
- GPIO_OUTPUT_TYPE
- GPIO_NUMBER
- GPIO_CR_MODE_INPUT
- GPIO_CR_CNF_ANALOG
- GPIO_CR_CNF_INPUT_FLOATING
- GPIO_CR_CNF_INPUT_PU_PD
- GPIO_CR_CNF_GP_OUTPUT_PP
- GPIO_CR_CNF_GP_OUTPUT_OD
- GPIO_CR_CNF_AF_OUTPUT_PP
- GPIO_CR_CNF_AF_OUTPUT_OD
- HAL_GPIO_Init(GPIO_TypeDef*, GPIO_InitTypeDef*) : void
- HAL_GPIO_DeInit(GPIO_TypeDef*, uint32_t) : void
- HAL_GPIO_ReadPin(GPIO_TypeDef*, uint16_t) : GPIO_PinState
- HAL_GPIO_WritePin(GPIO_TypeDef*, uint16_t, GPIO_PinState) : v
- HAL_GPIO_TogglePin(GPIO_TypeDef*, uint16_t) : void
- HAL_GPIO_LockPin(GPIO_TypeDef*, uint16_t) : HAL_StatusTypeD
- HAL_GPIO_EXTI_IRQHandler(uint16_t) : void
- HAL_GPIO_EXTI_Callback(uint16_t) : void

Problems | Tasks | Console | Properties

CDT Build Console [LED_Blink]

```
arm-none-eabi-objdump -h -S LED_Blink.elf  > "LED_Blink.list"
   text    data     bss     dec     hex filename
   5536      12    1644    7192    1c18 LED_Blink.elf
Finished building: default.size.stdout

Finished building: LED_Blink.list


13:12:34 Build Finished. 0 errors, 0 warnings. (took 555ms)
```

Build Analyzer | Static Stack Analyzer | Cyclomatic Complexity

LED_Blink.elf - /LED_Blink/Debug - Feb 7, 2025, 1:12:33 PM

Memory Regions | Memory Details

| Region | Start address | End address | Size | Free | Used | Usage (%) |
|---|---|---|---|---|---|---|
| RAM | 0x20000000 | 0x20004fff | 20 KB | 18.38 KB | 1.62 KB | 8.09% |
| FLASH | 0x08000000 | 0x0801ffff | 128 KB | 122.58 KB | 5.42 KB | 4.23% |

# STM32CudeIDE Project

## >> LED_Blink – stm32f1xx_hal.c

MicroProcessor - LED_Blink/Drivers/STM32F1xx_HAL_Driver/Src/stm32f1xx_hal.c - STM32CubeIDE

File Edit Source Refactor Navigate Search Project Run Window Help Hello Hyunwoo

Project Explorer

- LED_Blink
  - Binaries
  - Includes
  - Core
  - Drivers
    - CMSIS
    - STM32F1xx_HAL_Driver
      - Inc
      - Src
        - stm32f1xx_hal_cortex.c
        - stm32f1xx_hal_dma.c
        - stm32f1xx_hal_exti.c
        - stm32f1xx_hal_flash_ex.c
        - stm32f1xx_hal_flash.c
        - stm32f1xx_hal_gpio_ex.c
        - stm32f1xx_hal_gpio.c
        - stm32f1xx_hal_pwr.c
        - stm32f1xx_hal_rcc_ex.c
        - stm32f1xx_hal_rcc.c
        - stm32f1xx_hal_tim_ex.c
        - stm32f1xx_hal_tim.c
        - stm32f1xx_hal_uart.c
        - stm32f1xx_hal.c
      - LICENSE.txt
  - Debug
  - LED_Blink.ioc
  - STM32F103RBTX_FLASH.ld

LED_Blink.ioc | main.c | stm32f1xx_hal_gpio.c | stm32f1xx_hal.c

```c
363   * @note In the default implementation , SysTick timer is the source of time base.
364   *       It is used to generate interrupts at regular time intervals where uwTick
365   *       is incremented.
366   * @note This function is declared as __weak to be overwritten in case of other
367   *       implementations in user file.
368   * @param Delay specifies the delay time length, in milliseconds.
369   * @retval None
370   */
371 __weak void HAL_Delay(uint32_t Delay)
372 {
373   uint32_t tickstart = HAL_GetTick();
374   uint32_t wait = Delay;
375
376   /* Add a freq to guarantee minimum wait */
377   if (wait < HAL_MAX_DELAY)
378   {
379     wait += (uint32_t)(uwTickFreq);
380   }
381
382   while ((HAL_GetTick() - tickstart) < wait)
383   {
384   }
385 }
386
387 /**
388   * @brief Suspend Tick increment.
389   * @note In the default implementation , SysTick timer is the source of time base. It is
390   *       used to generate interrupts at regular time intervals. Once HAL_SuspendTick()
391   *       is called, the SysTick interrupt will be disabled and so Tick increment
392   *       is suspended.
393   * @note This function is declared as __weak to be overwritten in case of other
394   *       implementations in user file.
395   * @retval None
396   */
397 __weak void HAL_SuspendTick(void)
398 {
399   /* Disable SysTick Interrupt */
400   CLEAR_BIT(SysTick->CTRL, SysTick_CTRL_TICKINT_Msk);
401 }
402
```

**Millisecond 단위의 지연 시간 발생 함수**

Outline | Build Targets

- stm32f1xx_hal.h
- __STM32F1xx_HAL_VERSION_MAIN
- __STM32F1xx_HAL_VERSION_SUB1
- __STM32F1xx_HAL_VERSION_SUB2
- __STM32F1xx_HAL_VERSION_RC
- __STM32F1xx_HAL_VERSION
- IDCODE_DEVID_MASK
- uwTick : volatile uint32_t
- uwTickPrio : uint32_t
- uwTickFreq : HAL_TickFreqTypeDef
- HAL_Init(void) : HAL_StatusTypeDef
- HAL_DeInit(void) : HAL_StatusTypeDef
- HAL_MspInit(void) : void
- HAL_MspDeInit(void) : void
- HAL_InitTick(uint32_t) : HAL_StatusTypeDef
- HAL_IncTick(void) : void
- HAL_GetTick(void) : uint32_t
- HAL_GetTickPrio(void) : uint32_t
- HAL_SetTickFreq(HAL_TickFreqTypeDef) : HAL_StatusTypeDef
- HAL_GetTickFreq(void) : HAL_TickFreqTypeDef
- HAL_Delay(uint32_t) : void
- HAL_SuspendTick(void) : void
- HAL_ResumeTick(void) : void
- HAL_GetHalVersion(void) : uint32_t
- HAL_GetREVID(void) : uint32_t
- HAL_GetDEVID(void) : uint32_t
- HAL_GetUIDw0(void) : uint32_t
- HAL_GetUIDw1(void) : uint32_t
- HAL_GetUIDw2(void) : uint32_t
- HAL_DBGMCU_EnableDBGSleepMode(void) : void
- HAL_DBGMCU_DisableDBGSleepMode(void) : void
- HAL_DBGMCU_EnableDBGStopMode(void) : void
- HAL_DBGMCU_DisableDBGStopMode(void) : void
- HAL_DBGMCU_EnableDBGStandbyMode(void) : void
- HAL_DBGMCU_DisableDBGStandbyMode(void) : void

Problems | Tasks | Console | Properties

CDT Build Console [LED_Blink]

```
arm-none-eabi-objdump -h -S LED_Blink.elf  > "LED_Blink.list"
   text    data     bss     dec     hex filename
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Finished building: default.size.stdout

Finished building: LED_Blink.list


13:12:34 Build Finished. 0 errors, 0 warnings. (took 555ms)
```

Build Analyzer | Static Stack Analyzer | Cyclomatic Complexity

LED_Blink.elf - /LED_Blink/Debug - Feb 7, 2025, 1:12:33 PM

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| FLASH | 0x08000000 | 0x0801ffff | 128 KB | 122.58 KB | 5.42 KB | 4.23% |

# STM32CudeIDE Project

## >> LED_Blink – main.c

MicroProcessor - LED_Blink/Core/Src/main.c - STM32CubeIDE

File Edit Source Refactor Navigate Search Project Run Window Help Hello Hyunwoo

Project Explorer

- LED_Blink
  - Binaries
  - Includes
  - Core
    - Inc
      - main.h
      - stm32f1xx_hal_conf.h
      - stm32f1xx_it.h
    - Src
      - main.c
      - stm32f1xx_hal_msp.c
      - stm32f1xx_it.c
      - syscalls.c
      - sysmem.c
      - system_stm32f1xx.c
    - Startup
  - Drivers
    - CMSIS
    - STM32F1xx_HAL_Driver
      - Inc
      - Src
        - stm32f1xx_hal_cortex.c
        - stm32f1xx_hal_dma.c
        - stm32f1xx_hal_exti.c
        - stm32f1xx_hal_flash_ex.c
        - stm32f1xx_hal_flash.c
        - stm32f1xx_hal_gpio_ex.c
        - stm32f1xx_hal_gpio.c
        - stm32f1xx_hal_pwr.c
        - stm32f1xx_hal_rcc_ex.c
        - stm32f1xx_hal_rcc.c
        - stm32f1xx_hal_tim_ex.c
        - stm32f1xx_hal_tim.c
        - stm32f1xx_hal_uart.c
        - stm32f1xx_hal.c
      - LICENSE.txt
  - Debug
  - LED_Blink.ioc [Code Generation is required]
  - STM32F103RBTX_FLASH.ld

LED_Blink.ioc | main.c | main.h | stm32f1xx_hal_gpio.c | stm32f1xx_hal.c

```c
  */
int main(void)
{

  /* USER CODE BEGIN 1 */

  /* USER CODE END 1 */

  /* MCU Configuration--------------------------------------------------------*/

  /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
  HAL_Init();

  /* USER CODE BEGIN Init */

  /* USER CODE END Init */

  /* Configure the system clock */
  SystemClock_Config();

  /* USER CODE BEGIN SysInit */

  /* USER CODE END SysInit */

  /* Initialize all configured peripherals */
  MX_GPIO_Init();
  MX_USART2_UART_Init();
  /* USER CODE BEGIN 2 */

  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */
    HAL_GPIO_TogglePin(LED_GPIO_Port, LD2_Pin);
    HAL_Delay(500);
    /* USER CODE BEGIN 3 */
  }
```

Outline | Build Targets

- main.h
- huart2 : UART_HandleTypeDef
- SystemClock_Config(void) : void
- MX_GPIO_Init(void) : void
- MX_USART2_UART_Init(void) : void
- main(void) : int
- SystemClock_Config(void) : void
- MX_USART2_UART_Init(void) : void
- MX_GPIO_Init(void) : void
- Error_Handler(void) : void
- assert_failed(uint8_t*, uint32_t) : void

Problems | Tasks | Console | Properties

CDT Build Console [LED_Blink]

```
arm-none-eabi-objdump -h -S LED_Blink.elf  > "LED_Blink.list"
   text    data     bss     dec     hex filename
   5536      12    1644    7192    1c18 LED_Blink.elf
Finished building: default.size.stdout

Finished building: LED_Blink.list


13:12:34 Build Finished. 0 errors, 0 warnings. (took 555ms)
```

Build Analyzer | Static Stack Analyzer | Cyclomatic Complexity

LED_Blink.elf - /LED_Blink/Debug - Feb 7, 2025, 1:12:33 PM

Memory Regions | Memory Details

| Region | Start address | End address | Size | Free | Used | Usage (%) |
|---|---|---|---|---|---|---|
| RAM | 0x20000000 | 0x20004fff | 20 KB | 18.38 KB | 1.62 KB | 8.09% |
| FLASH | 0x08000000 | 0x0801ffff | 128 KB | 122.58 KB | 5.42 KB | 4.23% |

Writable | Smart Insert | 101 : 50 : 2721

# STM32CudeIDE Project

>> LED_Blink

- 프로젝트 생성시 CubeMX에서 C 코드 생성을 진행
- /* USER CODE BEGIN(END) XXX */
  코드의 각 부분 마다 User code begin/end로 구역이 구분되어 있음. 해당 구역 내에 코드를 작성을 권장
- User Code 주석 사이에 코드를 작성시, CubeMX에서 프로세서 설정을 수정하고 C 코드 생성을 진행할때 작성한 코드가 그대로 남음
- 개발 과정에서 여러 수정사항을 반영하면서 소스코드 작성이 가능함

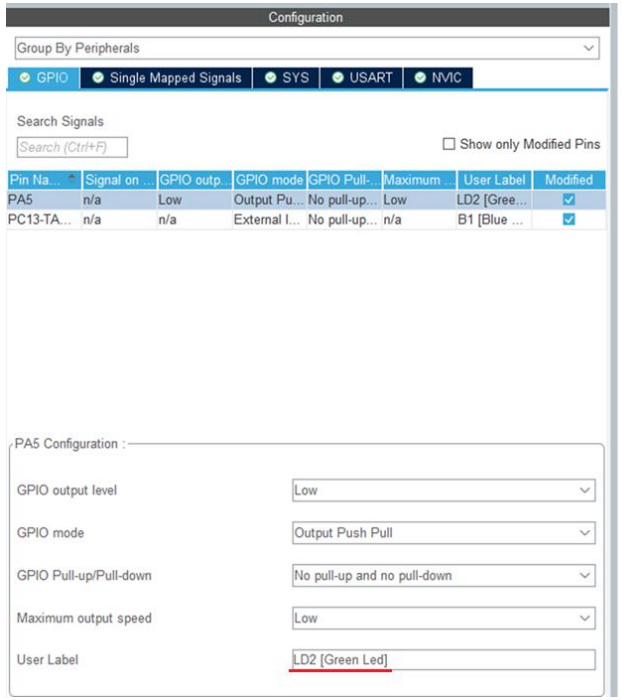
```
  /* Initialize all configured peripherals */
  MX_GPIO_Init();
  MX_USART2_UART_Init();
  /* USER CODE BEGIN 2 */

  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
      HAL_GPIO_TogglePin(LD2_GPIO_Port, LD2_Pin);
      HAL_Delay(500);
  }
  /* USER CODE END 3 */
}

/**
  * @brief System Clock Configuration
  * @retval None
  */
void SystemClock_Config(void)
```

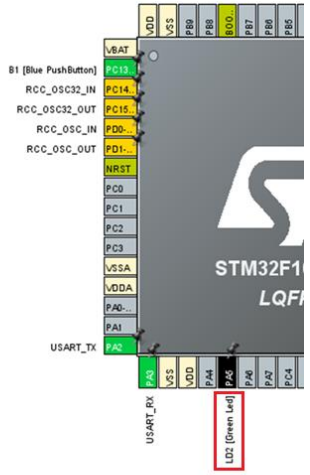
# STM32CudeIDE Project

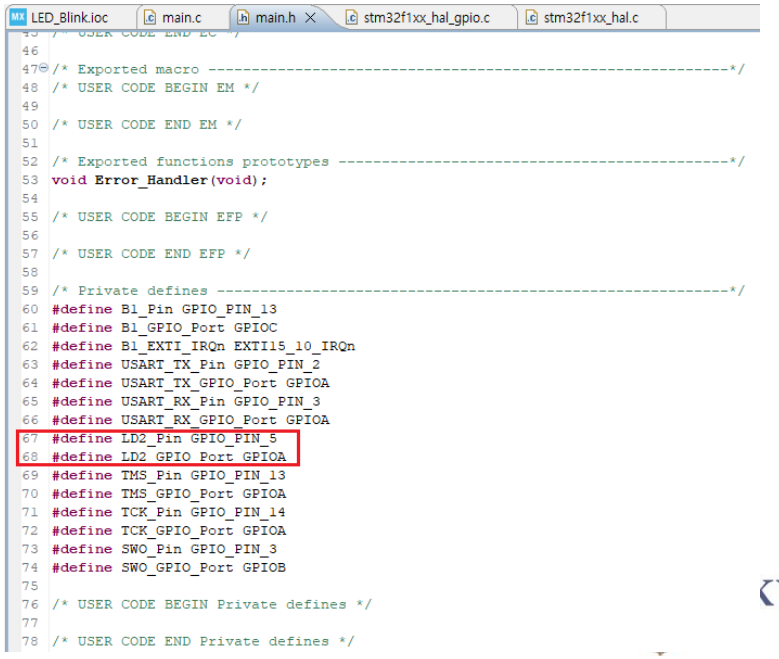
## >> LED_Blink

- CubeMX 상에서 핀에 대한 라벨링 가능
- 라벨은 main.h에서 해당 핀 선언을 통해 반영됨
- 따라서 Port A, Pin 5에 대한 상수 선언이 main.h에서 아래와 같이 변경됨
  LD2_Pin -> GPIO_PIN_5
  LD2_GPIO_Port -> GPIOA
- main.c에서 선언된 상수를 통해 보다 효율적으로 핀에 대한 제어가 가능함

```
  /* USER CODE BEGIN 3 */
    HAL_GPIO_TogglePin(LD2_GPIO_Port, LD2_Pin);
    HAL_Delay(500);
}
/* USER CODE END 3 */
```



# STM32CudeIDE Project

## >> LED_Blink

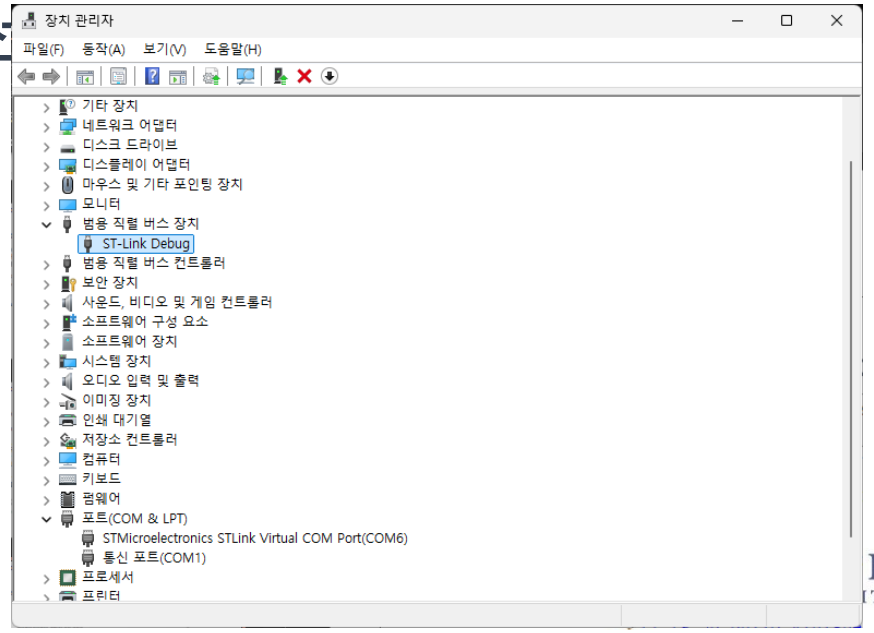
- 앞에서 진행하였던 Build Project 진행 (Ctrl + B 단축키)

Problems | Tasks | Console | Properties

CDT Build Console [LED_Blink]

```
17:19:50 **** Incremental Build of configuration Debug for project LED_Blink ****
make -j12 all
arm-none-eabi-size  LED_Blink.elf
   text    data     bss     dec     hex filename
   5680      12    1644    7336    1ca8 LED_Blink.elf
Finished building: default.size.stdout


17:19:50 Build Finished. 0 errors, 0 warnings. (took 259ms)
```

- 업로드에 앞서 보드 연결 확인 (Win + x 단축키, 장치 관리자 선




- ST-Link Debug와 STMicroelectronics STLink~ 확인

# STM32CudeIDE Project

## >> LED_Blink

- 앞에서 진행하였던 Build Project 진행 (Ctrl + B 단축키)

Problems | Tasks | Console | Properties

CDT Build Console [LED_Blink]

```
17:19:50 **** Incremental Build of configuration Debug for project LED_Blink ****
make -j12 all
arm-none-eabi-size  LED_Blink.elf
   text    data     bss     dec     hex filename
   5680      12    1644    7336    1ca8 LED_Blink.elf
Finished building: default.size.stdout


17:19:50 Build Finished. 0 errors, 0 warnings. (took 259ms)
```

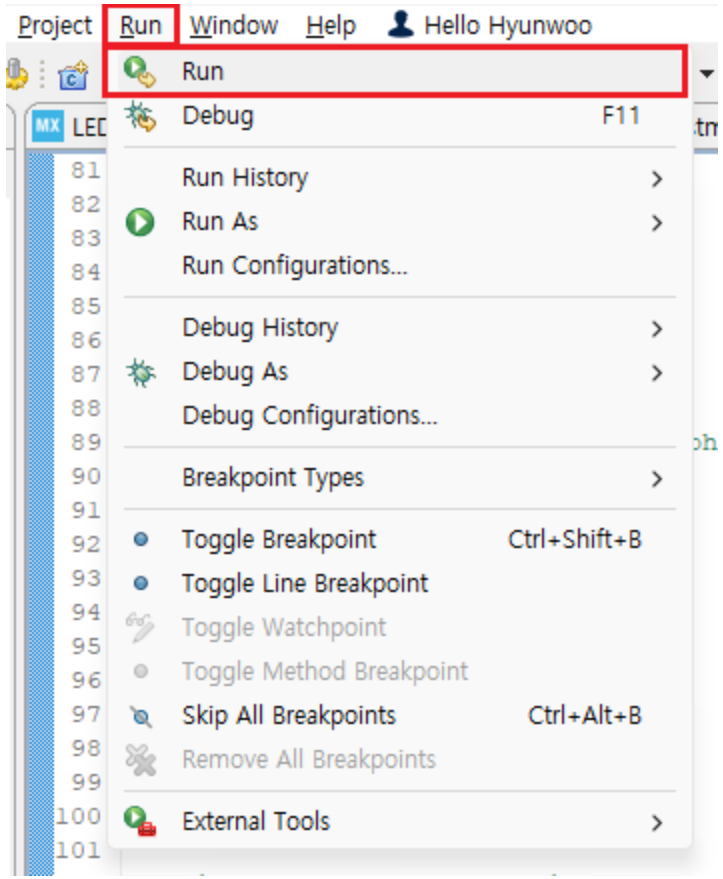
- 업로드에 앞서 보드 연결 확인 (Win + x 단축키, 장치 관리자 선택)
  ST-Link Debug와 STMicroelectronics STLink~ 확인

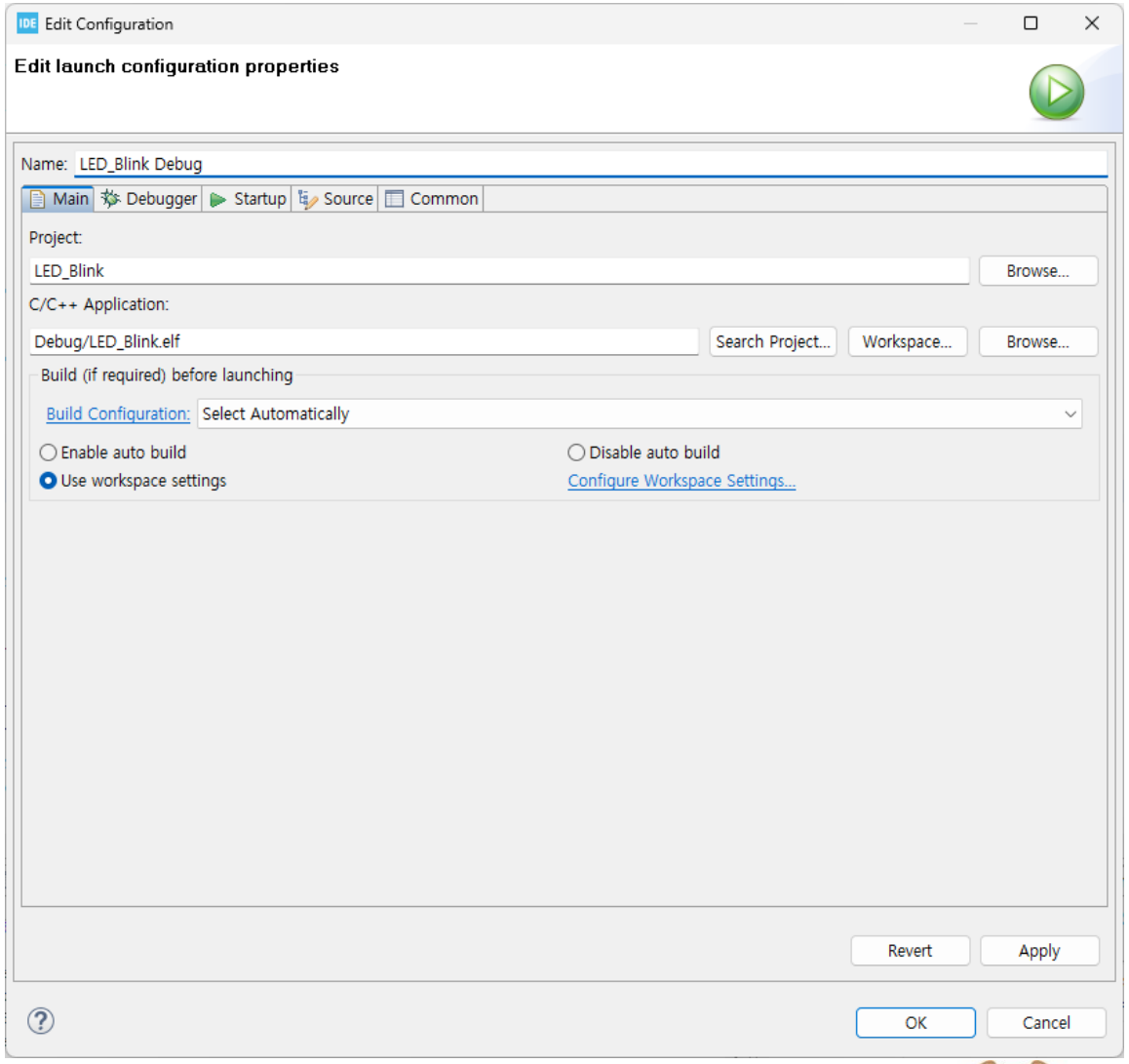
범용 직렬 버스 장치
- ST-Link Debug

포트(COM & LPT)
- STMicroelectronics STLink Virtual COM Port(COM6)
- 통신 포트(COM1)

# STM32CudeIDE Project

## >> LED_Blink – Upload & Run

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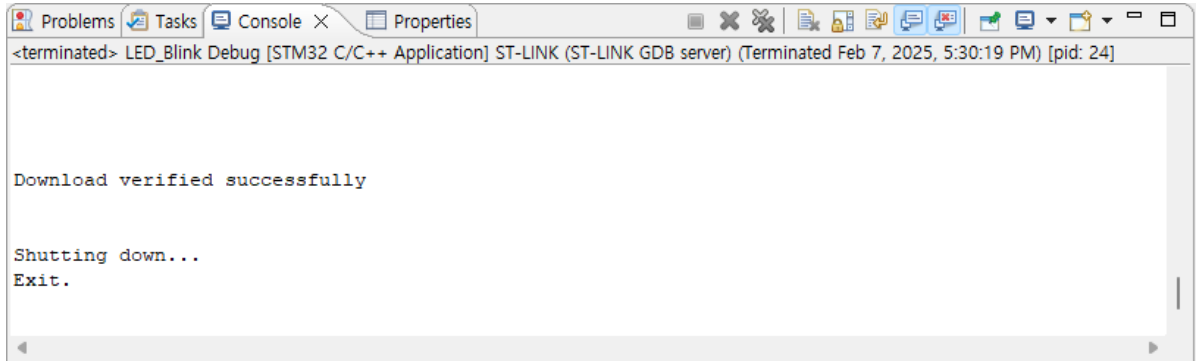
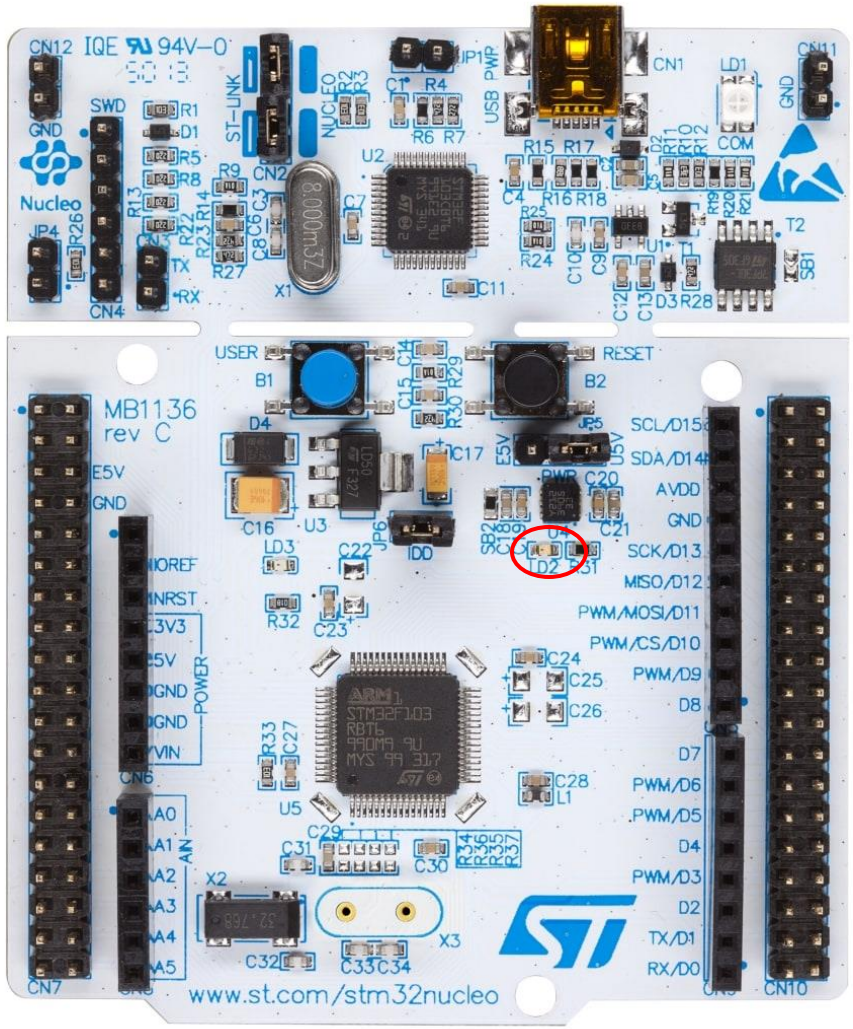
# STM32CudeIDE Project

## >> LED_Blink – Upload & Run

- 업로드 진행 완료 확인




- 보드 상의 LD2 LED가 1초에 한번씩 점등되는것을 확인



# STM32CudeIDE Project

## >> LED_Blink – GDB Debug

MP_2025 - LED_Blink/Core/Src/main.c - STM32CubeIDE

File Edit Source Refactor Navigate Search Project Run Window Help

Run
Debug F11
Run History
Run As
Run Configurations...
Debug History
Debug As
Debug Configurations...
Breakpoint Types
Toggle Breakpoint Ctrl+Shift+B
Toggle Line Breakpoint
Toggle Watchpoint
Toggle Method Breakpoint
Skip All Breakpoints Ctrl+Alt+B
Remove All Breakpoints
External Tools

Project Explorer
- LED_Blink
  - Binaries
  - Includes
  - Core
  - Drivers
  - Debug
  - LED_Blink.ioc
  - LED_Blink Debug.launch
  - STM32F103RBTX_FLASH.ld

```
99    {
100     /* USER CODE END WHILE */
101
102     /* USER CODE BEGIN 3 */
103       HAL_GPIO_TogglePin(LD2_GPIO_Port, LD2_Pin);
104       HAL_Delay(500);
105   }
106   /* USER CODE END 3 */
107 }
108
109 /**
110   * @brief System Clock Configuration
111   * @retval None
112   */
113 void SystemClock_Config(void)
114 {
115   RCC_OscInitTypeDef RCC_OscInitStruct = {0};
116   RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};
117
118   /** Initializes the RCC Oscillators according to the specified parameters
```

Confirm Perspective Switch

This kind of launch is configured to open the Debug perspective when it suspends.

This Debug perspective supports application debugging by providing views for displaying the debug stack, variables and breakpoints.

Switch to this perspective?

☐ Remember my decision

Switch No

Problems Tasks Console Properties

<terminated> LED_Blink Debug [STM32 C/C++ Application] ST-LINK (ST-LINK GDB server) (Terminated Feb 9, 2025, 10:31:30 AM) [pid: 17]

```
verifying ...

Download verified successfully

Shutting down...
Exit.
```

Build Analyzer Static Stack Analyzer Cyclomatic Complexity

LED_Blink.elf - /LED_Blink/Debug - Feb 9, 2025, 10:02:10 AM

Memory Regions Memory Details

| Region | Start address | End address | Size | Free | Used | Usage (%) |
|---|---|---|---|---|---|---|
| RAM | 0x20000000 | 0x20004fff | 20 KB | 18.38 KB | 1.62 KB | 8.09% |
| FLASH | 0x08000000 | 0x0801ffff | 128 KB | 122.58 KB | 5.42 KB | 4.23% |

Writable Smart Insert 103 : 50 : 2752

# STM32CudeIDE Project

## >> LED_Blink – GDB Debug

MP_2025 - LED_Blink/Core/Src/main.c - STM32CubeIDE

File Edit Source Refactor Navigate Search Project Run Window Help

Debug | Project Explorer

- LED_Blink Debug [STM32 C/C++ Application]
  - LED_Blink.elf [cores: 0]
    - Thread #1 [main] 1 [core: 0] (Suspended : Breakpoint)
      - main() at main.c:76 0x8000150
  - arm-none-eabi-gdb (13.2.90.20230627)
  - ST-LINK (ST-LINK GDB server)

LED_Blink.ioc | main.c | main.h

```
63   * @brief  The application entry point.
64   * @retval int
65   */
66 int main(void)
67 {
68
69   /* USER CODE BEGIN 1 */
70
71   /* USER CODE END 1 */
72
73   /* MCU Configuration--------------------------------------------------------*/
74
75   /* Reset of all peripherals, Initializes the Flash interface and the Systick. */
76   HAL_Init();
77
78   /* USER CODE BEGIN Init */
79
80   /* USER CODE END Init */
81
82   /* Configure the system clock */
83   SystemClock_Config();
84
85   /* USER CODE BEGIN SysInit */
86
87   /* USER CODE END SysInit */
88
89   /* Initialize all configured peripherals */
90   MX_GPIO_Init();
91   MX_USART2_UART_Init();
92   /* USER CODE BEGIN 2 */
93
94   /* USER CODE END 2 */
95
96   /* Infinite loop */
97   /* USER CODE BEGIN WHILE */
98   while (1)
99   {
100    /* USER CODE END WHILE */
101
102    /* USER CODE BEGIN 3 */
```

Variabl | Breakp | Express | Registe | Live Ex | SFRs

| Name | Type | Value |
|---|---|---|

Console | Problems | Executables | Debugger Console | Memory

LED_Blink Debug [STM32 C/C++ Application] [pid: 22]

```
Verifying ...

Download verified successfully
```

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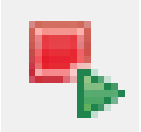
# STM32CudeIDE Project

## >> LED_Blink – GDB Debug

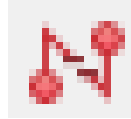
재시작


Debug 모드 종료 후 재시작

재개(F8), Breakpoint 지점까지 실행

일시정지

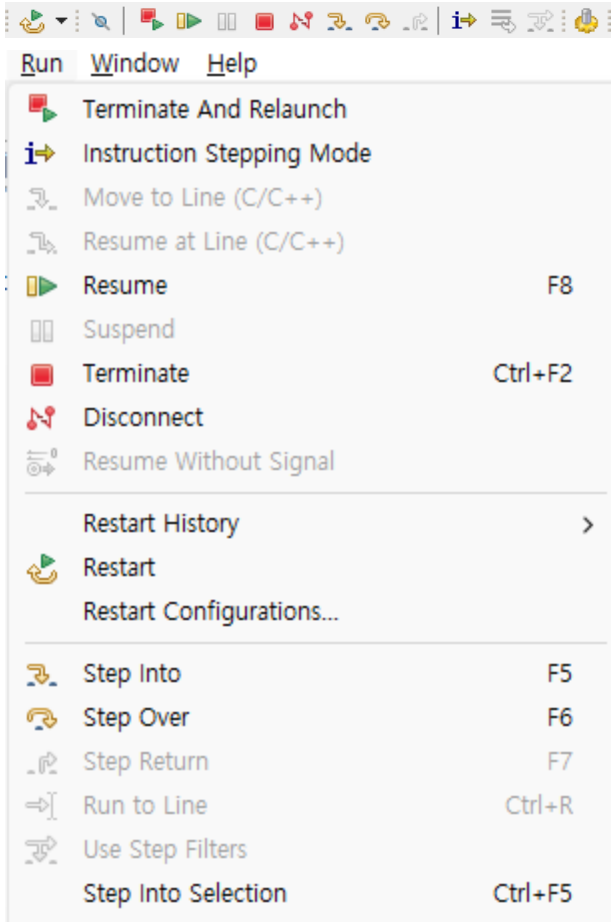
Debug 종료(Ctrl + F2)


연결 종료

Step into(F5) – 코드 한 줄 실행, 함수인 경우 함수 내부로 이동

Step over(F6) – 코드 한 줄 실행, 함수 호출 및 종료 후 다음 줄로 이동

Step Return(F7) – 함수 내부인 경우, 함수의 마지막까지 실행 후 호출된 지점으로 복귀

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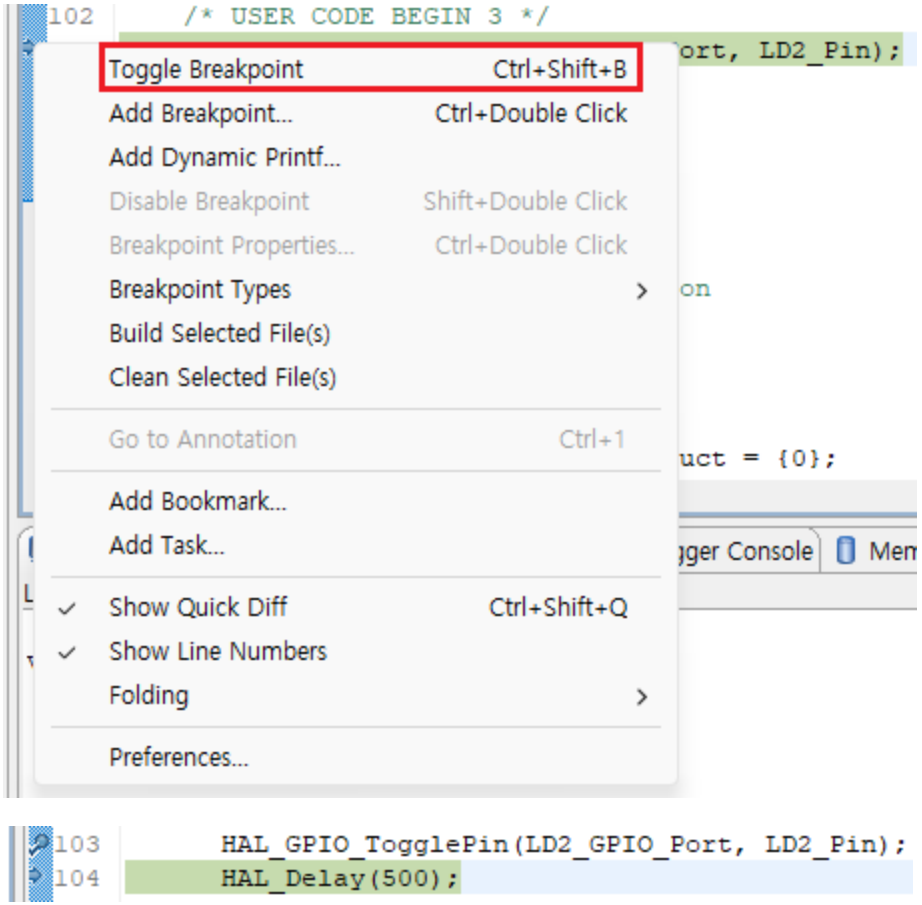
# STM32CudeIDE Project

## >> LED_Blink – GDB Debug

- Breakpoint, 중단점
  Toggle Breakpoint 혹은 Add Breakpoint를 통해 중단점 설정

- 중단점을 설정하고 재개 버튼을 누르면 코드가 순차적으로 실행되나, 중단점에서 코드 실행이 멈추는 것을 확인

- 특정 지점에서 코드 실행 흐름을 검토하거나, 변수의 현재 값, 함수 호출, 메모리 상태 등 코드의 실행 상태를 점검할 수 있음



# STM32CudeIDE Project

## >> LED_Blink – GDB Debug

LED_Blink.ioc | main.c | main.h | startup_stm32f103rbtx.s | stm32f1xx_hal.c

```c
  MX_GPIO_Init();
  MX_USART2_UART_Init();
  /* USER CODE BEGIN 2 */

  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */
  while (1)
  {
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
      HAL_GPIO_TogglePin(LD2_GPIO_Port, LD2_Pin);
      HAL_Delay(local_val);
      if (local_val > 500)
          local_bool = !local_bool;
      my_function(&local_val, local_bool);
  }
  /* USER CODE END 3 */
}
```

Variabl | Breakp | Express | Registe | Live Ex | SFRs

| Name | Type | Value |
|---|---|---|
| local_val | int | 7 |
| local_bool | _Bool | true |

실시간으로 변수를 추적하는 예시

LED_Blink.ioc | main.c | main.h | startup_stm32f103rbtx.s | stm32f1xx_hal.c

```c
  GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
  HAL_GPIO_Init(LD2_GPIO_Port, &GPIO_InitStruct);

  /* EXTI interrupt init*/
  HAL_NVIC_SetPriority(EXTI15_10_IRQn, 0, 0);
  HAL_NVIC_EnableIRQ(EXTI15_10_IRQn);

/* USER CODE BEGIN MX_GPIO_Init_2 */
/* USER CODE END MX_GPIO_Init_2 */
}

/* USER CODE BEGIN 4 */
void my_function(int *input, bool up_down) {
    if (up_down)
        (*input)++;
    else
        (*input)--;

}
/* USER CODE END 4 */

/**
```

Variabl | Breakp | Express | Registe | Live Ex | SFRs

| Name | Type | Value |
|---|---|---|
| input | int * | 0x20004ff0 |
| up_down | _Bool | true |

실시간으로 함수 내부를 확인 하는 예시

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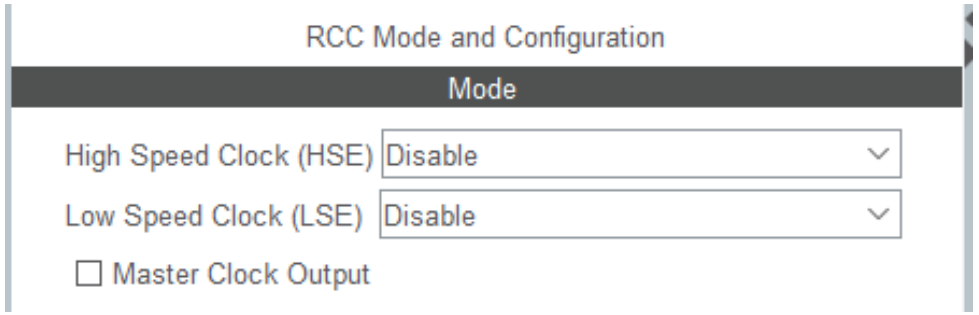
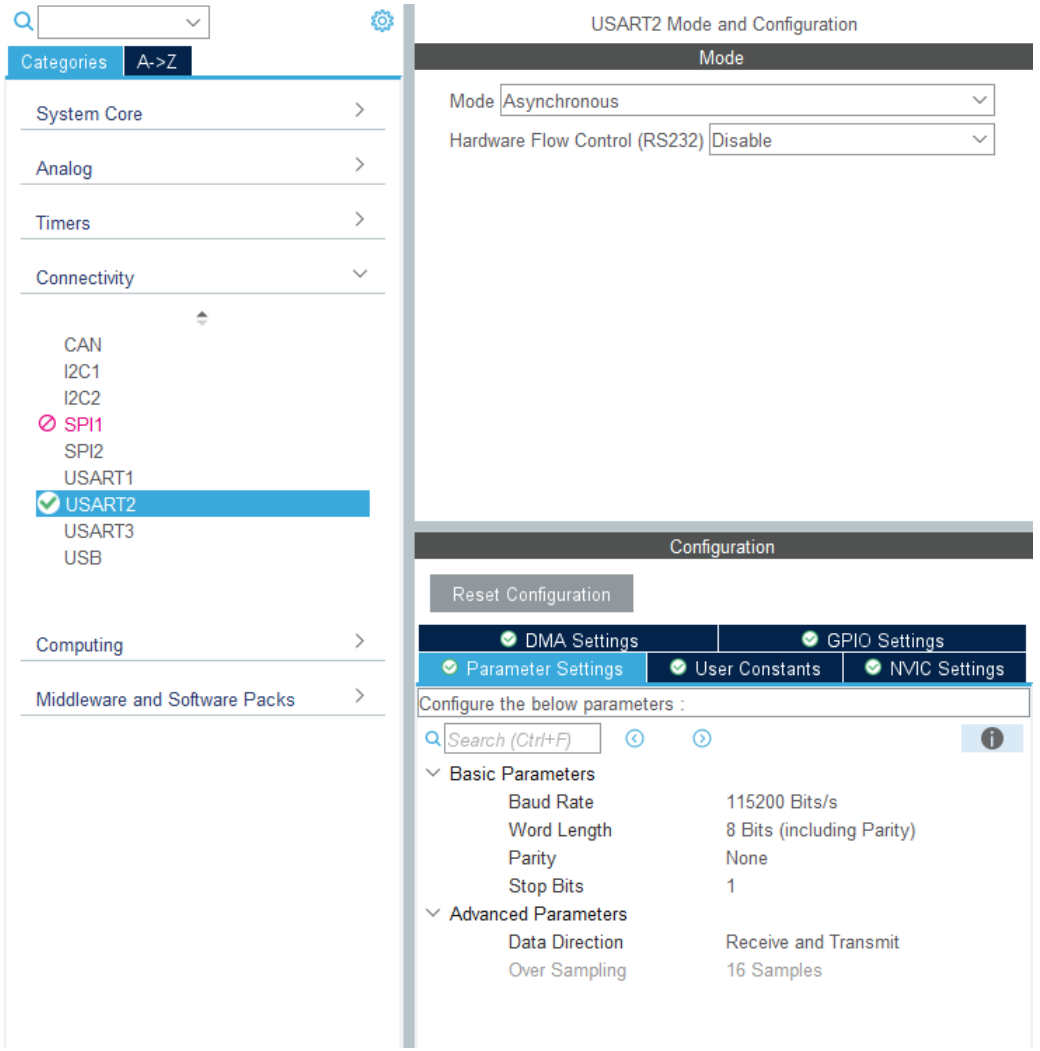
# STM32CudeIDE Project

## >> LED_Blink – Serial Debug with printf

- 이전 예제와 동일하게 RCC 설정
- USART 설정 확인
- 프로젝트 생성 (Project -> Generate Code)
- 프로젝트 빌드 (Ctrl + B)



# STM32CudeIDE Project

## >> LED_Blink – Serial Debug with printf

- **main.c**

```c
/* Private includes ----------------------------------------------------------*/
/* USER CODE BEGIN Includes */
#include <stdio.h>
/* USER CODE END Includes */
```

```c
/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */
#ifdef __GNUC__
#define PUTCHAR_PROTOTYPE int __io_putchar(int ch)
#else
#define PUTCHAR_PROTOTYPE int fputc(int ch, FILE *f)
#endif

PUTCHAR_PROTOTYPE
{
    if (ch == '\n')
        HAL_UART_Transmit (&huart2, (uint8_t*) "\r", 1, 0xFFFF);
    HAL_UART_Transmit (&huart2, (uint8_t*) &ch, 1, 0xFFFF);

    return ch;
}
```

```c
  while (1)
  {
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
      printf("Hello World!\n");
      HAL_Delay(1000);
  }
  /* USER CODE END 3 */
```

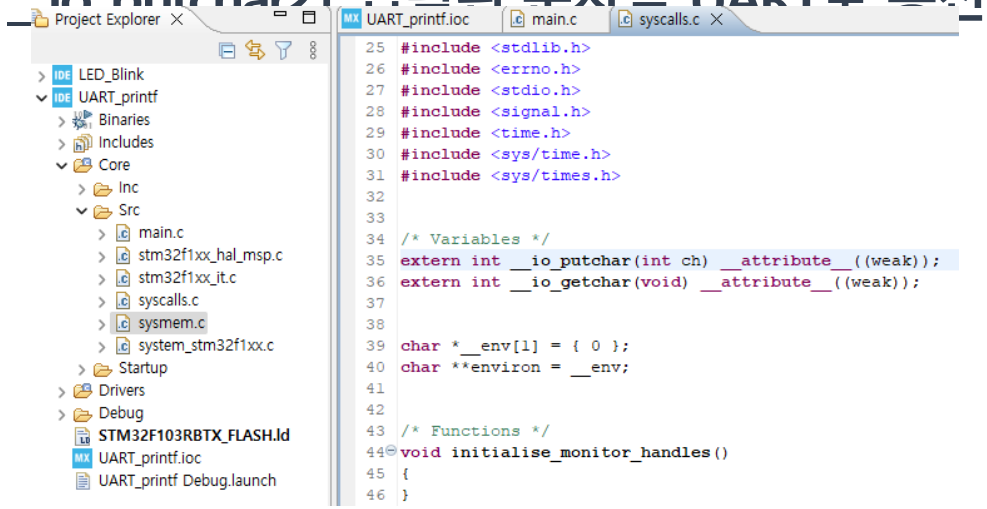
# STM32CudeIDE Project

## >> LED_Blink – Serial Debug with printf

- syscalls.c를 보면 __io_putchar가 weak로 선언되어 제공됨
  사용자가 해당 함수를 원하는대로 재정의 가능함
  weak 심볼은 동일한 이름으로 강한 심볼이 존재할 경우, 강한 심볼을 사용(즉 main.c에서 선언한 함수로 대체)

- main.c에서 매크로 선언을 통해 __io_putchar 재정의(GNU GCC 툴체인 사용)

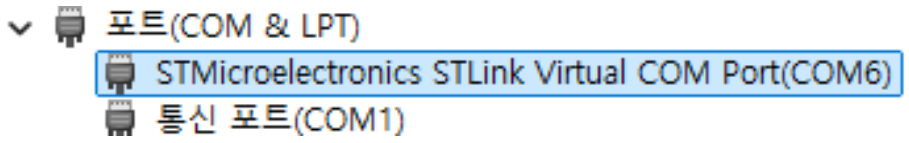
- __io_putchar가 입력된 문자를 UART로 통신하도록 재정의



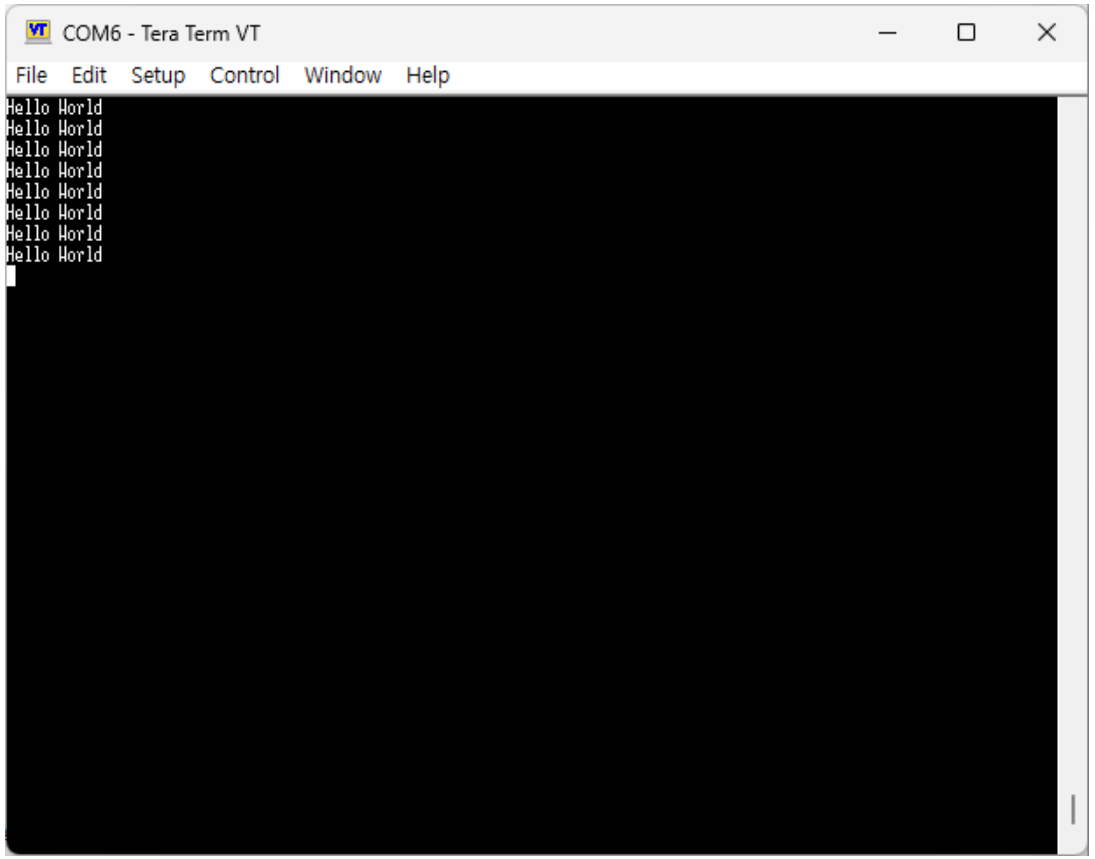

```
/* Private user code ---------------------------------------------------------*/
/* USER CODE BEGIN 0 */
#ifdef __GNUC__
#define PUTCHAR_PROTOTYPE int __io_putchar(int ch)
#else
#define PUTCHAR_PROTOTYPE int fputc(int ch, FILE *f)
#endif

PUTCHAR_PROTOTYPE
{
    if (ch == '\n')
        HAL_UART_Transmit (&huart2, (uint8_t*) "\r", 1, 0xFFFF);
    HAL_UART_Transmit (&huart2, (uint8_t*) &ch, 1, 0xFFFF);

    return ch;
}
```

# STM32CudeIDE Project

## >> LED_Blink – Serial Debug with printf




- Nucleo보드가 연결된 포트를 확인, Tera Term에서 포트 설정(Baud rate 115200, CubeMX 설정값)
- printf를 통해 Serial로 “Hello World”가 출력됨
- 기존의 C/C++, python과 같이 파일입출력 방식을 통해 문자, 변수 등 현재 상태를 나타내어 디버깅할 수 있음
- 가장 직관적이고 단순한 디버깅 방법, 터미널과 USB-Serial 만으로 간단하게 디버깅 환경 구축이 가능함
- GDB와 달리 런타임에서 디버깅 하기때문에 세부적인 부분에 대한 추적이 불가능함



# Thank you

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Human-Robot Interaction Laboratory

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